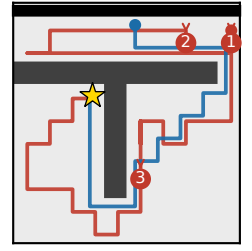
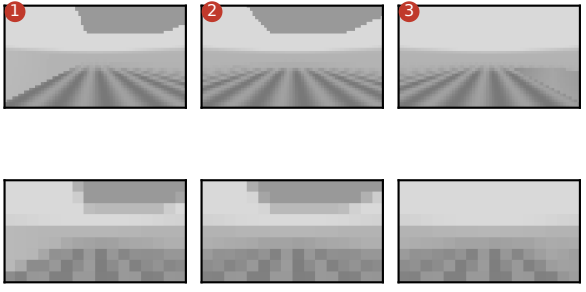
a Layout 3, real paths **b Agent view at 1–3: camera (top), 16×9 input (bottom)** **c Inputs**





receives

- pooled camera (144)
- contact, last reward
- own last action (4)
- frozen visual pose belief (102; true cell 100% of 39k native decisions)

never receives

- goal location
- layout or wall map
- trial / phase boundaries

4 allocentric moves
45 decisions per trial
reward only at the goal